

Caption-Guided Cross-Attention Fusion for Remote Sensing Land-Cover Classification and Segmentation

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Abstract—We compare four cross-modal fusion strategies for caption-guided remote sensing scene understanding on a 10,024-image subset of ARAS400K. A frozen RemoteCLIP-ViT-B/32 backbone is paired with late, FiLM, gated, and cross-attention (CA) fusion modules over five LLM-generated caption types, and the same modules are applied to patch-level multi-label classification and segmentation. CA wins every head-to-head against late fusion across captions, seeds, and tasks, reaching mAP 0.948 ± 0.003 and mIoU 0.711 ± 0.003 . Gated fusion underperforms late concatenation in both tasks, a negative result we report. We further conduct three ablations. A threshold sensitivity sweep ($\tau \in \{5, 10, 20\}\%$) shows the fusion gain over the image-only baseline grows monotonically with τ , from $+0.097$ to $+0.153$ mAP. A full 5×4 segmentation matrix confirms the fusion ranking at scale. A backbone-capacity ablation to RemoteCLIP-ViT-L/14 decreases segmentation mIoU by 0.041, exposing a frozen-backbone limit that motivates parameter-efficient fine-tuning as future work. All 500+ training runs and code are public.

I. INTRODUCTION

Remote sensing (RS) imagery contains heterogeneous land cover at many spatial scales. Recent vision-language models (VLMs) generate captions that describe each scene in natural language, raising a practical question: can captions improve scene-level prediction beyond what an image-only model achieves? The answer depends on how the modalities are fused.

We study this question on a 10,024-image subset of ARAS400K [11] (the validation subset of the released 400,240-image corpus, comprising 100,240 real and 300,000 synthetic images), with 7-class segmentation masks, composition percentages, and five LLM-generated caption strategies (vision-only, text-only from composition, and hybrid). Captions for the same image can range from highly leaky (the hybrid and text-only streams encode composition percentages literally and these percentages also define our multi-label targets) to noisy and unreliable (vision-only LLM captions hallucinate on this subset, for example labelling a 92%-grass scene as “barren”). This spectrum tests how each fusion mechanism handles caption quality.

Research question: How effective is cross-attention multi-modal fusion compared to late fusion for RS scene understanding using ViTs and LLM-generated captions?

Contributions: (i) A four-way fusion comparison (late, FiLM, gated, CA) with three seeds across 5 captions and 2 tasks. (ii) A negative result: gated fusion underperforms late concatenation, the sigmoid gate collapses one modality. (iii) Three ablations: threshold sensitivity, a full 5×4 segmentation matrix, and a backbone-capacity ablation that severely affects segmentation results in an unexpected direction. (iv)

An example-samples figure that makes the inputs and outputs concrete on three diverse validation scenes. (v) Public code, 500+ tracked runs, and reproducible feature caches.

II. LITERATURE SURVEY

VLMs jointly represent visual and textual information through contrastive learning [1] and have gained traction in RS, where Wen et al. [2] identify cross-modal fusion as a major open challenge. RemoteCLIP [3], the first CLIP adapted to RS, outperforms vanilla CLIP by up to 6.39% on zero-shot classification via domain-adapted pretraining. Multimodal LLMs such as EarthGPT [4] and RingMoGPT [5] unify captioning, VQA, grounding, and change detection within a single instruction-tuned framework.

A central design choice is how to fuse the two modalities. Cross-modal attention has beaten unimodal methods on RS VQA [6] and visual grounding [7], while SegCLIP [8] fuses text with image features for RS segmentation. FiLM [9] offers an attention-free alternative through per-feature affine modulation. Few studies directly compare these mechanisms on RS scene-level tasks, and the role of caption-generation strategy on fusion performance remains under-explored despite evidence that caption source and quality matter [10].

III. METHODS

Task: An obvious formulation of the problem is 8-way dominant-cover classification at a 50% threshold (7 land-cover classes plus an eighth “no dominant” class for scenes where no class exceeds the threshold). This formulation turns out to be trivial under text input alone (mAP ≈ 0.83 from captions). We instead use 7-bit multi-label class presence at threshold τ , with $\tau = 10\%$ as the default. The same image patch features drive a per-patch segmentation head on a 7×7 grid (Path B).

Backbone: We use a frozen RemoteCLIP-ViT-B/32 backbone [3]. Per-token features are captured through forward hooks on the visual `ln_post` and text `ln_final` layers, after which we apply the model’s own projection per token. This gives 50 image tokens (CLS + 49 patches) and 77 text tokens at 512 dim.

Fusion modules: Four variants share a 256-unit ReLU–Dropout–Linear head. Late concatenates pooled image and text vectors. FiLM generates per-feature (γ, β) from pooled text and modulates the image CLS, with the modulator’s final layer zero-initialised so the model starts at the image-only baseline. Gated mixes the two pooled vectors through a sigmoid gate. Cross-Attention uses `nn.MultiheadAttention` with 8 heads. For classification, text tokens are queries over image patches as keys and values, mean-pooled before the head.

TABLE I
PATH A BEST RESULT PER FUSION FAMILY (3-SEED MEAN, $\tau = 10\%$, B/32 BACKBONE).

Fusion	Best mAP	Macro F1	Mean mAP
Image-only	0.817 ± 0.001	0.702 ± 0.006	—
Late	0.908 ± 0.003	0.836 ± 0.009	0.864
Gated	0.897 ± 0.001	0.811 ± 0.007	0.857
FiLM	0.943 ± 0.001	0.876 ± 0.008	0.903
Cross-Attn	0.948 ± 0.003	0.867 ± 0.014	0.908

For segmentation the direction is inverted: image patches are queries over text tokens as keys and values, producing per-patch logits. Only the fusion module and head are trained.

Training and evaluation: All conditions use the full 10,024-image subset with an 80/20 stratified split on dominant class. Each run uses AdamW (lr 10^{-3} , weight-decay 10^{-4} , 30 epochs, batch 256 for Path A and 128 for Path B) with three random seeds. Metrics are mAP and per-class F1 (Path A) and mIoU and per-class IoU (Path B). With features cached after a single backbone pass, the entire experimental matrix (≈ 500 runs, broken down as 117 main + 189 threshold + 63 full B/32 seg + 126 L/14) completes in ≈ 3 hours on one A100. All runs are public on Weights & Biases and the codebase is on GitHub [12], [13].

IV. RESULTS

A. Main results on the B/32 backbone

Path A classification: The best run is cross-attention with hybrid_qwen3-vl-8b at mAP 0.948 ± 0.003 versus an image-only baseline of 0.817 ± 0.001 (Fig. 2, Table I). CA beats late fusion on every caption (Δ mAP from +0.036 to +0.051). FiLM is the close runner-up at 0.005 mAP behind CA. Gated fusion underperforms even simple late concatenation, plausibly because the sigmoid gate collapses toward a near-unimodal representation, although direct inspection of gate values would be needed to confirm this.

Noise robustness: With the noisy vision_qwen3-vl-8b caption, late fusion gains nothing over image-only ($0.815 \approx 0.817$) while CA recovers +0.049 mAP. The pattern is consistent with CA being more robust to noisy captions than the parametric fusions, although token-level attention analysis would be needed to confirm whether unreliable tokens are explicitly down-weighted.

Per-class decomposition: Minor classes that the image-only baseline fails on (Shrub F1 0.00, Built-up 0.65, Barren 0.61) recover under CA (Shrub 0.66, Built-up 0.79, Barren 0.79). Captions and images carry complementary information on these classes.

Path B segmentation: On the full 5×4 B/32 matrix (Table II), the best run is CA + hybrid_qwen3-vl-8b at mIoU 0.711 ± 0.003 versus image-only 0.574 ± 0.003 . The Path A fusion ranking transfers exactly. The CA-vs-Late delta is positive on all five captions, the noise-robustness signature persists with vision-only captions, and per-class IoU lifts concentrate on minor classes (Shrub $0.05 \rightarrow 0.41$, Barren $0.39 \rightarrow 0.59$).

TABLE II
PATH B SEGMENTATION mIoU PER FUSION FAMILY ACROSS THE FIVE CAPTION STRATEGIES (3-SEED MEAN, B/32 BACKBONE, 7×7 PATCH GRID). IMAGE-ONLY BASELINE mIoU 0.574 ± 0.003 . BEST PER ROW IN BOLD.

Fusion	hyb-gem	hyb-qwn	txt-qwn	vis-gem	vis-qwn	Mean
Late	0.663	0.669	0.673	0.582	0.575	0.632
Gated	0.660	0.667	0.675	0.581	0.573	0.631
FiLM	0.689	0.694	0.693	0.615	0.613	0.661
Cross-Attn	0.709	0.711	0.701	0.608	0.610	0.668

B. Ablation study

Three ablations probe distinct dimensions of the research question: task definition (threshold), task scope (full segmentation matrix), and backbone capacity (B/32 versus L/14).

(A) Threshold sensitivity: We sweep $\tau \in \{5, 10, 20\}\%$ over the full 5×4 Path A matrix with 3 seeds (189 runs). The image-only baseline degrades as τ rises (mAP $0.841 \rightarrow 0.817 \rightarrow 0.791$) because higher thresholds drop minor positives. The CA-vs-baseline gap grows with τ : +0.097 at $\tau = 5\%$, +0.131 at $\tau = 10\%$, and +0.153 at $\tau = 20\%$. Fig. 3 shows the family-wise best mAP per τ . CA is best on average at every threshold. FiLM slightly exceeds CA at $\tau = 5\%$ (0.944 vs 0.938) but trails at $\tau = 10\%$ and $\tau = 20\%$. The default $\tau = 10\%$ is a balanced operating point and does not change the fusion ranking.

(B) Full segmentation matrix: A preliminary segmentation run covered only 7 conditions (3 captions, late and CA). We close the matrix to all 21 conditions (5×4 + image-only) with 3 seeds. The full matrix confirms the Path A ranking transfers to dense prediction without exception: CA wins on each of the 5 captions, FiLM is the consistent runner-up, and gated again falls below late on average (mean mIoU 0.631 vs 0.632). This is the most direct test that the headline findings were not artefacts of the small condition set.

(C) Backbone capacity (severely-affects ablation): We re-extract patch features with a larger RemoteCLIP-ViT-L/14 backbone (304M params vs 86M, 16×16 patches at 224 input vs 7×7 , feature dim 768 vs 512) and re-run both task matrices. The intent was to test whether higher patch resolution improves segmentation. Results contradict that hypothesis. On Path A, L/14 only marginally improves the best CA mAP ($0.948 \rightarrow 0.956$) and degrades the image-only baseline ($0.817 \rightarrow 0.803$). On Path B the effect is severe in the opposite direction: every fusion family loses mIoU under L/14, including the best CA run (mIoU $0.711 \rightarrow 0.670$, $\Delta = -0.041$). Fig. 4 contrasts the two backbones per fusion family, and Table III reports per-class IoU at L/14. Despite the drop, the fusion ranking and per-class lift pattern of L/14 mirror those of B/32 (Shrub $0.034 \rightarrow 0.378$, +0.344, and Barren $0.310 \rightarrow 0.511$, +0.200). The pattern is preserved. Only the absolute scale shifts.

V. DISCUSSION

Across these experiments, CA consistently outperforms late fusion across captions, seeds, thresholds, and tasks. On the Path A B/32 matrix the gain over the image-only baseline is +0.131 mAP. On the Path B B/32 matrix it is +0.137 mIoU. On the L/14 backbone the absolute scale shifts but the relative gain is preserved (Path B +0.137). The average family



Fig. 1. Representative validation scenes. Each row shows the input image, ground-truth mask, best-CA segmentation prediction, and the five LLM caption strategies (truncated). Predictions are shown at the L/14 backbone because the 16×16 patch grid is visually finer than the 7×7 B/32 grid. Quantitative results elsewhere use B/32 as the main setting. Top, a dominant-class scene (Crop 85%). Middle, a balanced multi-class scene (Tree, Shrub, Grass each $\geq 28\%$). Bottom, a forested scene with a notable minor-class component (Shrub 19%).

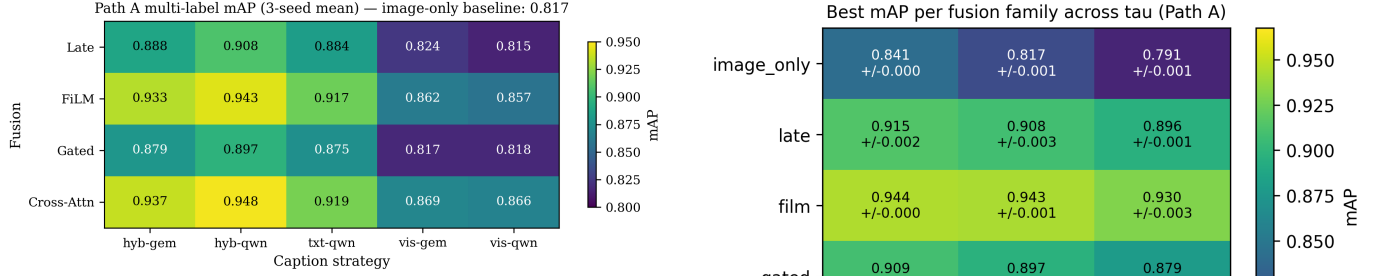


Fig. 2. Path A multi-label mAP at $\tau = 10\%$ (3-seed mean). Rows are fusion modules, columns are caption strategies. Image-only baseline: 0.817 ± 0.001 .

TABLE III
PER-CLASS IOU UNDER THE L/14 BACKBONE, IMAGE-ONLY VERSUS BEST CA (HYBRID_QWEN). FUSION-INDUCED LIFT IS LARGEST ON MINOR CLASSES.

Class	Image-only	Best CA	Δ
Tree	0.760	0.814	+0.054
Shrub	0.034	0.378	+0.344
Grass	0.715	0.793	+0.077
Crop	0.641	0.759	+0.118
Built-up	0.505	0.575	+0.070
Barren	0.310	0.511	+0.200
Water	0.764	0.858	+0.094
mIoU	0.533	0.670	+0.137

ranking is $CA \approx FiLM > Late \approx Gated$, with CA giving the strongest main results and the most reliable improvement over late fusion. FiLM is competitive throughout and slightly exceeds CA in the $\tau=5\%$ setting (0.944 vs 0.938 best mAP), so we do not claim that CA dominates FiLM unconditionally.

Comparison with the literature: Bazi et al. [6] report

Fig. 3. Threshold ablation. Best mAP per fusion family at $\tau \in \{5, 10, 20\}\%$ on Path A. Fusion gain over image-only is monotone in τ .

cross-attention beating unimodal methods on RS VQA, and Lan et al. [7] show the same on grounding. Our result is consistent with these findings on a third RS task type (scene-level multi-label classification) and on segmentation. SegCLIP [8] addresses RS segmentation with a related text-conditioned design. We use it as methodological context rather than a numerical baseline, since its datasets, class taxonomies, supervision, and evaluation protocols differ from ours. The vanilla-CLIP versus RemoteCLIP gap (avg. +0.031 mAP for RemoteCLIP in an earlier backbone-source ablation we ran on the same matrix) is qualitatively consistent with the 6.39% zero-shot gap reported by Liu et al. [3], even though their evaluation protocol is different.

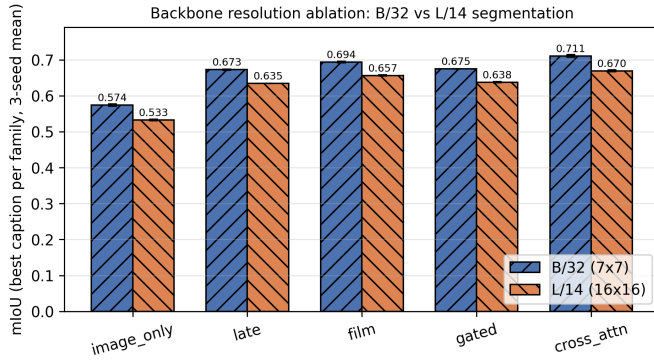


Fig. 4. Backbone ablation. Best mIoU per fusion family at B/32 (hatch //) and L/14 (hatch \\\). The larger, higher-resolution L/14 backbone reduces segmentation mIoU across all fusion families.

Why does L/14 regress on segmentation? The L/14 backbone has $3.5\times$ more parameters and $5.2\times$ more patch tokens, but the trainable head and the training budget are unchanged. More spatial tokens through the same cross-entropy loss make per-patch optimisation noisier, and the L/14 image embedding is trained for image-level alignment rather than patch-level discrimination. The image-only seg baseline already drops by 0.041 before any fusion is added, supporting this reading. Partial fine-tuning, for instance LoRA on the last attention blocks, is a natural next experiment to test whether this gap can be recovered.

VI. CONCLUSION

We compared four cross-modal fusion strategies for caption-guided RS land-cover classification and segmentation on a 10,024-image subset of ARAS400K with a frozen RemoteCLIP backbone. Cross-attention consistently outperformed late fusion across 5 captions, 3 seeds, two tasks, two backbones, and three thresholds, and gave the strongest main result (mAP 0.948 ± 0.003 on Path A, mIoU 0.711 ± 0.003 on Path B). FiLM was competitive throughout and slightly exceeded CA at $\tau=5\%$. Threshold sensitivity showed the fusion gain over the image-only baseline grows monotonically with τ . The full segmentation matrix replicated the Path A pattern at scale. A backbone-capacity ablation to ViT-L/14 produced a 0.041 mIoU drop on segmentation, revealing a limit of the frozen-backbone paradigm. Because the hybrid and text-only captions are derived from the same composition statistics that define our multi-label targets, results in those settings should be read as oracle-text upper bounds rather than fully independent language supervision. The vision-only-caption results provide the cleaner test of caption-guided fusion. Parameter-efficient fine-tuning of the backbone is a natural future direction. The full codebase, ≈ 500 tracked runs, and reproducible feature caches are public.

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